PROJECT TITLE: PAHINGA: Development of an Integrated Resort Management System for AML's Resort in Poras, Boac, Marinduque

Keywords: *Resort Booking*

Proponent(s): 1. *Harvey M. Lunar - Lead Backend Developer*

2. *Louise Angela Grace Regencia – QA*

3. *Tereinz Motol - DevOps*

4. James Ashley Largo - Frontend Developer

CHAPTER 1: TECHNICAL DESCRIPTION & PROPOSAL

1.1 Background of the Study

AML's Resort is a private hospitality business located in Poras, Boac, Marinduque, offering accommodation and recreational facilities such as pools, convention rooms, and cottages. As a service provider for tourists and residents, the resort requires an efficient system to manage its operations.

Currently, the resort uses a manual-based system for handling reservations, payments, attendance, and report generation. This leads to errors, delays, and difficulty in managing records. Additionally, the resort lacks a proper system for monitoring maintenance activities, which may result in overlooked facility issues.

To address these problems, the proposed PAHINGA (Integrated Resort Management System) will be developed. It includes modules for Reservation & Availability, Billing & Payment, Attendance Monitoring, Maintenance Management, and Reports Management. The system aims to improve efficiency, accuracy, and overall service quality by automating resort operations.

1.2 Objectives (General and Specific)

- General Objective:

  To design, develop, and deploy a web-based Integrated Resort Management System using the MERN stack for AML's Resort that enables efficient online booking and administrative management by the end of the semester.

- Specific Objectives:

  1. To implement a secure user authentication system using JWT and bcrypt, separating access between customers and administrators.

2. To develop a RESTful API that performs CRUD operations for users, bookings, and resort services.

3. To create an interactive frontend interface using React.js that allows users to view resort details and book online.

4. To integrate a MongoDB database for efficient storage and retrieval of booking and user data.

5.To deploy the system using cloud platforms (Vercel/Render) with minimal to zero critical errors.

1.3 Review of Related Literature and Systems

- Related Related Article 1: [Ismail, S. 2023. Resort management system using laravel framework (PHP). BRAC University Institutional Repository. (Jan. 2023)]

- By replacing a physical, manual bottleneck with a centralized digital hub, this system solves the main problem. The system changes the "guarantee" of a reservation from a verbal promise to a database record, so a guest doesn't have to go to the beach just to get a room. The system's biggest problems are that it depends completely on the internet and that data could be wrong if staff don't digitize walk-ins right away. Also, it is still open to double-booking and mistakes made when entering data by hand because it doesn't work with third-party booking sites.

- Related Article 2: [Ramento, L.F. et al. 2025. Web-Based Reservation and Management System for Nature Spring Resort. Journal of Pure and Applied Sciences. 4, 2 (Apr. 2025).]

- This system replaces inefficient manual booking with a centralized, real-time digital interface, which eliminates human error and the need for physical visits. It makes reservations less of a hassle by automating guest management and tracking availability. This gives management quick access to accurate data. But its usefulness is still limited because it relies completely on internet access and staff still have to manually digitize walk-in bookings to keep the database accurate. If the system isn't integrated with third-party booking platforms, it could also have operational "gaps" where outside reservations conflict with the internal, digitized schedule.

- Related Article 3: [Sambherao, S. V. et al. 2025. Resort Management System. International Journal of Research Publication and Reviews. (Apr. 2025)]

- This system replaces separate manual tasks with a single digital hub that lets bookings, inventory, and billing sync up in real time. This gets rid of data silos and makes it easier to make decisions, but the system is still vulnerable to network failures that could stop operations. Also, its wide range makes it hard to learn, and if staff training and strong conflict resolution aren't done regularly, the complexity may cancel out the efficiency gains that were meant to happen.

- Related Article 4: [Islam A. 2020. Resort Management System. United International University. (Feb. 2020)]

- This system takes the guesswork out of travel by showing guests real-time room availability and prices. This way, they don't have to worry about getting to a resort that is already full. Management can keep an eye on profitability and use customer feedback to make operations better by digitizing the reservation process and tracking finances. The system does have some problems with reputation management, though. For example, the review module makes the resort open to public scrutiny, and if maintenance costs or inventory changes aren't updated manually, the data can be wrong. In the end, the system only works well if the internet is always available, which isn't always the case in remote tourist areas.

- Related Article 5: [Kabir, A. et al. 2022. Hotel Management System. EWU Institutional Repository. (Oct. 2022)]

- The Hotel Management System gets rid of the problems with keeping records by hand by replacing physical ledgers with a centralized Oracle database. This makes it possible to get data right away, make updates without any problems, and verify users' identities. This moves to digitally eliminate paper waste and speeds up processes like billing and registration by a huge amount. However, the system's use of a HTML front-end may make real-time interactivity less possible, and its performance depends entirely on having stable power and network connections, which makes it more likely to go down than a physical register.

- Related Article 6: [Betrand, C. U. et al. 2023. Automated Hotel Management Information System. Journal of Research in Engineering and Computer Sciences. (Apr. 2023)]

- The automated Hotel Management System uses PHP and JavaScript to bring together different parts of the business, like bookings, cancellations, and employee management, into a modular, searchable digital space. This makes it easier to find and manage information. Using the SSADM methodology makes the transition more organized and less likely to lose data. It also automates tedious tasks like making reports and bills. The system is limited by its "simplified scope," which means it only has a few core modules instead of a full suite of hotel services. It also relies completely on technical uptime, which means that if the server or power goes down, the whole thing could stop working.

- Related Article 7: [Aryadivani, N. M. N. et al. 2025. Design and Development of a Website-Based Reservation Information System at the Backyard Villa. Jurnal Galaksi. 2, 2 (Aug. 2025)]

- The web-based system fixes the villa's manual problems by giving guests a self-service booking platform that gets rid of the delays and lost data that come with phone calls and

text messages. It makes sure that information is always up-to-date and that reports are done in a professional way by centralizing management and guest access. But the system has some problems. For one, the Waterfall development method is too strict, which makes future updates hard. For another, the system relies completely on internet connectivity, which means that the business could stop working if the service goes down.

- Related Article 8: [Lee Jia Mei and Norfaradilla Wahid 2023. Development of a Web-based Reservation Management System for Walters Farm. Applied Information Technology and Computer Science. 4, 2 (Nov. 2023)]

- The Walters Farm system uses PHP Laravel and MySQL to automate ticket sales and centralize bookings, which cuts down on the need for staff to do things by hand. The Prototype model made sure that users were happy by using iterative design, but the system is still limited by the fact that it needs hardware that works with QR codes and the possibility of architectural instability if the prototype wasn't fully hardened for large-scale tourism spikes.

- Related Article 9: [Ng, C.C. and Mohd Zaki Mohd Salikon 2022. Hotel Reservation Management System. Applied Information Technology and Computer Science. 3, 2 (Nov. 2022).]

- This online Hotel Reservation Management System gets rid of the problems that come with manual booking bottlenecks by giving customers a single place to check availability in real time and book rooms directly without any extra fees. Following the Waterfall Model guarantees that the system is well-organized and fully functional, making it easier for employees to handle both online and in-person reservations. This cuts down on the amount of work they must do. But the system is limited because of the "inflexibility of its development methodology," which means that once a phase is done, it's hard to make changes. Also, it only works if there is continuous internet access and system maintenance, which means that technical problems or being offline could slow down the booking process and make the customer experience worse.

- Related Articles 10: [Fayaz, Muhammad and Khan, Hamza. 2022. Online Hotel and Restaurant Management System (2022).]

- The computerized system solves the burden of manual "file work" by using a PHP and MySQL framework to automate room tracking, financial reporting, and guest management, allowing a small team to handle a rapidly growing customer base. By providing a reliable, web-accessible interface, it ensures data transparency and eliminates the physical effort of sorting through paper records. However, the system is limited by its external hosting dependency on Host Pakistan, meaning a server or internet failure could halt operations. Additionally, while it increases digital efficiency, it

cannot substitute for the physical staff shortages required to maintain food and service quality as the business continues to expand.

- Related Article 11: [Bemile R. et al. 2014. Online Hotel Reservation System. International Journal of Innovative Science, Engineering & Technology. (2014).]

- The online reservation system solves the problem of having to book in person by letting guests book rooms from anywhere with a "virtual tour" and a digital selection process. This means that guests don't have to send someone else to book for them or arrive early. By making the reservation official through a website, it gives the traveler a reliable guarantee of a place to stay. A major **limit**, though, is that there isn't enough information about how to integrate or sync payments in real time with other booking platforms. This could lead to overbooking if manual updates don't keep up with digital reservations.

- Related Article 12: [De Silva, P. B. N. N. 2017. Web Site & Web Based Hotel Reservation Management System for Sky Lodge Hotel. University of Colombo School of Computing. (2017).]

- The Web-Based Hotel Management System Fixes Sky Lodge's problems by putting all of its manual tasks, like managing rooms, making reservations, and keeping track of expenses, on one digital platform. The system improves the accuracy of calculations, cuts down on the time employees spend on reservations, and expands its international marketing reach with a dedicated website by replacing paper-based transactions with a PHP and MySQL-driven interface. One big problem, though, is that there are no automated external integrations. Without real-time currency conversion for foreign guests or automated synchronization with global travel agents, the system may still need to be watched over by a person to handle international financial issues and avoid double-bookings.

- Related Article 13: [Deeti, Sumanth; Manne, Sravani; and Vundavalli, Veerendra, "Hotel Management System" (2016).]

- The Global Hotel Management System makes it easier to manage hotels that are spread out over a large area by bringing together room availability and guest registrations into a single JSP and Servlet-based platform. It replaces manual coordination with automated business validations and a tiered access model (Admin/User). This keeps data consistent and lets guests book units from afar based on real-time schedules. A major limitation, though, is that it uses old architecture (JSP/Servlets) and doesn't have built-in payment gateways or automated check-in features. This could make it less efficient in a modern, high-traffic global market.

- Related Article 14: [Khairiah Abd Aziz and Suhaila Mohd Yasin 2021. MyHomestay Application System for Booking and Approval. Applied Information Technology and Computer Science. 2, 2 (Nov. 2021).]

- The MyHomestay Application fixes the problem of high commission fees and scattered information by giving the Kota Tinggi district a low-cost, localized platform. Built with Android Studio and Firebase, it makes homestay descriptions more consistent to help tourists make decisions. It also gives rural owners a separate digital channel that they didn't have before, when they had to rely on unreliable social media posts or costly third-party apps. However, a major drawback is that it only works on Android devices. Since it was made on Windows 10, it can't be used by iOS users, which could limit the number of travelers it can reach and the number of homestay owners it can help.

- Related Article 15: [Chidi U. B. et al. 2023. Automated Hotel Management Information System. Journal of Research in Engineering and Computer Sciences. (2023).]

- The Automated Hotel Reservation Management System makes it easier to handle complicated hospitality tasks by breaking them down into smaller, more manageable parts, such as managing floors, tracking employees, and handling money. Built with PHP and JavaScript, it improves operational efficiency by automating check-ins, cancellations, and report generation. It also uses the SSADM methodology to make sure that the transition from old processes goes smoothly. However, a major limitation is that the project itself admits that it has a narrow scope. The system only includes a "few modules" for educational purposes, so it doesn't have all the features needed for full-scale hotel operations and would need a lot of extra parts to work in a professional setting.

1.4 Methodology & Solution

The study will use the Agile Development Methodology, which supports continuous improvement and iterative development of the system. The proposed system will be developed using the MERN Stack, consisting of MongoDB for database management, Express.js for the backend framework, React.js for the frontend user interface, and Node.js for server-side execution. In terms of system architecture, the frontend built with React will communicate with the backend through a REST API. The backend, powered by Node.js and Express.js, will handle client requests and interact with MongoDB to store, retrieve, and manage data efficiently. For testing, the system will implement a comprehensive strategy that includes unit testing using Jest to validate individual components and functions, integration testing using Supertest to ensure proper API endpoint functionality, and end-to-end testing using Postman and Newman to verify complete user workflows.

1.5 Target Users / Beneficiaries

- **Beneficiary 1: Customers (Guests) -** Customers or guests are the primary users of the system. They will use the platform to view resort information, check rooms or cottage availability, and make reservations online. The system allows them to book conveniently without the need for walk-ins or manual inquiries. It also enables faster transactions and provides a smoother and more efficient booking experience.
- **Beneficiary 2: Resort Employees (Staff) -** Resort employees will use the system to manage day-to-day operations such as handling bookings, assisting customers, and updating reservation statuses. The system also automates processes like recording customer details and tracking services, which helps reduce manual work, minimize errors, and improve overall efficiency in operations.
- **Beneficiary 3: System Administrators (Management) -** System administrators or resort management will have full control over the system. They are responsible for managing bookings, updating resort information, monitoring customer data, and overseeing overall system operations. Admins can also generate reports, manage user accounts, and make strategic decisions based on accurate and real-time data provided by the system.
- **Validation / Certification -** To ensure the validity and reliability of the system, the proposed project will be evaluated and validated by the management of AML's Resort or a qualified IT expert. A certification or validation document will be secured to confirm that the system meets the operational needs and standards of the resort.

1.6 Significance of the Study

The implementation of the PAHINGA system marks a significant shift from time-consuming and cumbersome manual recording processes to a more automated and efficient digital system. By addressing the key problem areas experienced by AML's Resort, the study contributes to improved accuracy, consistency, and accountability in managing resort data such as bookings, customer information, and transactions.

This project is important because it enhances the overall operational efficiency of the resort. It reduces human errors, minimizes delays in reservation handling, and simplifies administrative tasks, allowing staff to focus more on delivering quality service. For customers, the system provides convenience through faster booking procedures, real-time availability, and smoother transactions, resulting in better overall experience.

In terms of financial benefits, the system can help increase reservations by making the resort accessible online and improving customer satisfaction, which may lead to repeat bookings. It also reduces operational costs by lessening the need for manual labor and paperwork.

Furthermore, the study serves as a benchmark for other small and local resorts in the area by demonstrating how digital transformation can improve service quality and operational

management. It highlights the importance of adopting modern technologies to remain competitive in the growing hospitality industry.

CHAPTER 2: MINIMUM VIABLE PRODUCT (MVP) DOCUMENTATION

2.1 System Architecture

[Provide a textual description (and optionally an embedded diagram) showing how your React Frontend, Node/Express Backend, and MongoDB database interact over the internet.]

2.2 Tech Stack & Tools

- Frontend: React.js, Vite, React Router
- Backend: Node.js, Express.js
- Database: MongoDB Atlas, Mongoose
- Testing: Jest, Supertest, Postman, Newman
- Deployment: Vercel / Render

2.3 Database Schema (Data Modeling)

- **Users:** Fields include full Name, email, password, role, and department. The role Enum includes ADMIN, FRONTDESK, EVENTS, and FACILITY. Passwords are securely hashed before storage.
- **Reservations:** Fields include bookingNumber, roomType, guestName, checkInDate, checkOutDate, status, and handledBy. The handledBy field automatically references the full name of the staff member who processed the booking.
- **Services:** Fields include serviceName, category, qtyRequested, qtyDelivered, balance, reservation, facility, and recordedBy. The qtyRequested and qtyDelivered fields track the number of services requested and completed for each reservation, respectively.
- **Facilities:** Fields include facilityName, location, coordinates, type, capacity, status, and photos. This collection connects to the facility field in the Services module, specifically linking facilityName, type, and capacity for monitoring and reporting purposes.

CHAPTER 3: REST API DESIGN DOCUMENTATION

3.1 Overview & Intent

- Base URL: pahinga-1.onrender.com
- Purpose: The API provides endpoints for authentication, guest management, reservation tracking, facility monitoring, service reconciliation, event scheduling, file uploads, and report generation within the Integrated Resort Management System.

3.2 Authentication & Authorization

- Strategy: JSON Web Tokens (JWT) passed via authentication. Upon a successful guest or staff login, the server issues a token that must be included in all secured API requests through the authorization header.

3.3 Endpoint Documentation

Endpoint Name: *User Registration*

- Method & URL: [POST] https://pahinga-1.onrender.com/api/auth/register
- Authorization Required: No
- Request Format (Headers & Body):

```
JSON

 {

"username": "newstaff",

"password": "password123",

"name": "Juan Dela Cruz",

"role": "employee”

}

Success Response:


HTTP 201

{

"message": "User registered successfully",

"user": {

"username": "newstaff",
```

```
"role": "employee",

"name": "Juan Dela Cruz"

}

}
```

End Point Name: User Login
Method & URL: [POST] https://pahinga-1.onrender.com/api/auth/login
Auth Required: No
Request Body:

```
json
{

"username": "admin",
"password": "admin123"

}

Success Response:

HTTP 200

{

"token": "jwt_token_here",
"user": {
"username": "admin",
"role": "admin",
"name": "Resort Administrator"

}

}
```

End Point Name: Get all Services
Method & URL: [GET] https://pahinga-1.onrender.com/api/services
Auth Required: No
Success Response:

HTTP 200

[

{

"_id": "service_id",
"id": "S001",
"name": "Swimming Pool",
"description": "Resort swimming pool access",
"price": 500,
"category": "activity"

}

]

EndPoint Name: Get All Bookings
Method & URL: [GET] https://pahinga-1.onrender.com/api/bookings
Auth Required: Yes (Bearer Token)
Headers:

Authorization: Bearer

Success Response:

HTTP 200

[

{

"_id": "booking_id",
"id": "BK001",
"customerName": "Juan Dela Cruz",
"email": "juan@example.com",
"service": "Swimming Pool",
"checkIn": "2024-01-20",
"checkOut": "2024-01-22",
"guests": 4,
"status": "confirmed"
}

]

EndPoint Name: Create Booking

Method & URL: [POST] https://pahinga-1.onrender.com/api/bookings

Auth Required: Yes (Bearer Token)

Request Body:

```
Json

{

"id": "BK002",
"customerName": "Maria Santos",
"email": "maria@example.com",
"phone": "09123456789",
"service": "Cottage",
"checkIn": "2024-02-15",
"checkOut": "2024-02-17",
"guests": 6,
"totalAmount": 3000

}
```

3.4 Advanced API Features

- Pagination/Filtering/Sorting: *The current version of the API primarily uses standard CRUD endpoints. Some modules may support filtering or lookup endpoints such as permit options, site search, or geospatial search depending on the implemented route.*
- Versioning Strategy: *The API uses /api/v1 in the route path to support future updates without breaking older client integrations. Future major changes may be introduced under /api/v2.*

3.5 Testing Tools & Examples

- Tools Used: Postman for manual testing, Newman for automated E2E testing.
- Sample URL/Fetch Example:

```JavaScript


fetch('https://pahinga-1.onrender.com/api/services', {

method: 'GET',

headers: {'Content-Type': 'application/json'

}
```

```
})

.then(response => response.json())

.then(data => console.log(data));
```

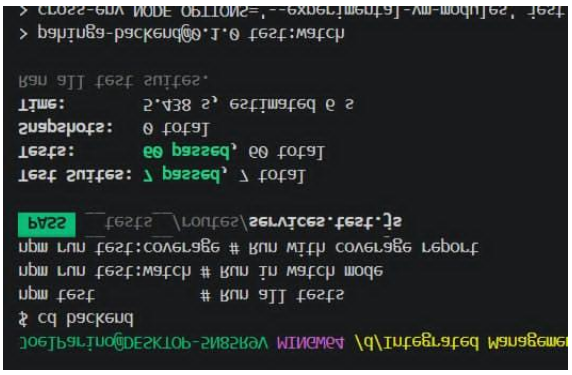
CHAPTER 4: DEPLOYMENT & LINKS

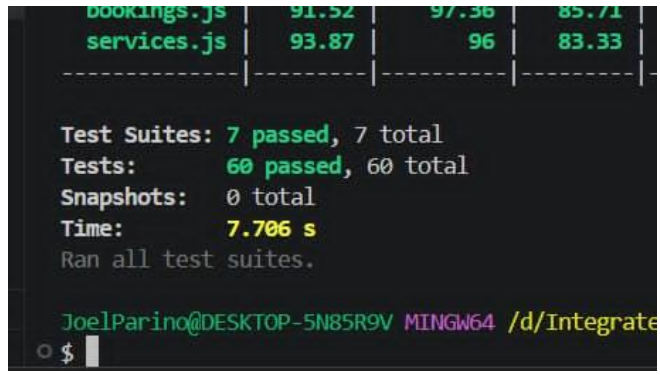
4.1 Live Application Links

- Frontend Web Application URL: https://pahinga-mu.vercel.app
- Backend API Base URL: https://pahinga-1.onrender.com
- GitHub Master Repository URL: https://github.com/joelp2002/Pahinga

4.2 Testing Coverage Summary

- Unit & Integration Tests:





- End-to-End (E2E) Tests: Postman was utilized to validate the entire workflow of the AMLS Resort Management System—from login to accessing secured modules. The tests covered admin authentication, staff account creation, and role-based access to

reservations, services, facilities, and reports, including restricted access for unauthorized roles. These validations confirmed that the system effectively implements JWT authentication and role-based access control across all deployed API endpoints.